



The type of T&M equipment required to perform these tests include network testers, cable antenna testers, MPEG codec testers, spectrum analysers and power meters. Oscilloscopes, bench-top digital multimeters (DMMs), spectrum analysers, signal sources and audio analysers are generally used for testing at R&D and manufacturing stages.

Profit margin depending on accuracy is high, but so is the competition among the companies producing T&M equipment. Some big names competing in this sector are Tektronix, Harris, Rohde & Schwarz, Keysight Technologies, Anritsu and Comcon Technologies.

The move to digital

"In India, analogue is the widely used form of transmission media. Now we are using orthogonal frequency-division multiplexing (OFDM) for modulation of data. This has led to an increase in the amount of data transmitted in the available bandwidth. Another welcome addition is the move to digital media," says Vishal Gupta, senior applications specialist (RF/MW) at Keysight. He adds, "Five years back, India's major mode of transmission was analogue, even in the studios. However, all broadcasters like public or private are moving ahead and introducing the latest standards. Going forward we are all moving towards digital technology. So the latest digital standards allow more video channels as well in the same frequency band and allow for better spectral efficiency. Equipment from five years back were focused on analogue and the latest are on digital."

The type of equipment ranges from simple signal analysers to complex multiple signal comparators with noise reduction and auto tuning. With this wide a range of equipment, these can be employed at different levels such as high-end equipment for R&D, hand-held equipment for testing and bench-

The testing process

The testing process for audio-video test equipment follows a certain pattern. The device under test (DUT) has to pass through some fixed parameters. In most of the test cases, the output of the DUT is compared to some standard reference signal.

A longer but accurate method is to test the signal upon reception by comparing it to the transmitted signal. Once the error signal is known, steps can be taken to minimise it, which would employ the reference signal. This method reduces the errors in the received signal and hence enables better broadcasting. It is more in tune with broadcasting systems, where the transmission and reception takes place in real-time. In such systems, signal quality is the important parameter.

Another more common method is to test the signal by passing it through the DUT and comparing the original signal and the output of the DUT. This method minimises the errors in the signal generated by the DUT. This method of testing is more common with audio-video system manufacturers. For them the important factor is the quality of equipment and hence the DUT can be tuned to best parameters.

top equipment for basic testing and maintenance.

Equipment for field

With high revenue generation, R&D for T&M equipment is also very high-level. New product releases in this field were not frequent but have been technically sound. The last decade has seen a change in this pattern and several new releases and modifications to previous systems are happening on a more frequent basis.

"Now we have specific equipment catering to audio-video testing including signal generators and analysers being used by the broadcast industry. Tools catering to radio testing are being introduced as well," says Gupta. He adds, "Today we have handheld tools in monitoring for field research equipped with GNSS, which provide signal conditions at different locations in real time."

Productivity enhancements

May 2015 saw DK-Technologies adding the DK T7 waveform monitor (WFM) to its portfolio. The T7 WFM is based on the T7 + + motherboard and runs software from DK's PTO800 video series. The unit operates in multi-channel metering, multi-set of loudness K-weighted relative to full scale (LKFS) loudness metering, signal test generator, all within a power consumption of 12 watts. "Essentially, it allows engi-

neers to work step-wise through the entire audio clip, rather than having to go back and measure loudness from the top for every tweak," says Kjems Hansen, product marketing director, DK-Technologies, in a press release. He continues, "The DK T7 was already the most comprehensive audio meter on the market, but with these new features it can deliver functionality that no other audio meter can offer."

Another move is the touch-screen interface for the equipment. This makes the equipment interactive and easier to handle, although now precision is further required in handling. For many measurements the established analyser displays and controls are familiar and efficient. However, the expanding functionality of signal analysers has led to a much deeper hard-key/soft-key menu structure and a dramatic expansion of display types.

In a latest addition to Keysight Technologies equipment, the X-series features a larger screen with multi-touch functionality similar to that of tablets. The measurement display and settings can be controlled by gestures such as single or double tapping to select a parameter or expand a window, dragging, pinching to zoom and scale displays, and press-and-hold to access context-sensitive selections. It includes an auto-tune feature to find and centre the likely signal of